

The Influence of Plant Root Systems on Subsurface Flow: Implications for Slope Stability

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Although research has explained how plant roots mechanically stabilize soils, in this article we explore how root systems create networks of preferential flow and thus influence water pressures in soils to trigger landslides. Root systems may alter subsurface flow: Hydrological mechanisms that promote lower pore-water pressures in soils are beneficial to slope stability, whereas those increasing pore pressure are adverse. Preferential flow of water occurs in the following types of root channels: (a) channels formed by dead or decaying roots, (b) channels formed by decayed roots that are newly occupied by living roots, and (c) channels formed around live roots. The architectural analysis of root systems improves our understanding of how roots grow initially, develop, die, and interconnect. Conceptual examples and case studies are presented to illustrate how root architecture and diverse traits (e.g., diameter, length, orientation, topology, sinuosity, decay rate) affect the creation of root channels and thus affect preferential flow.

Keywords: macropores, root channels, preferential flow paths, landslides, root architecture

A landslide is defined as the gravitational movement of a mass of soil or debris along a sliding surface at depth (Sidle and Ochiai 2006). It is well known that vegetation can play a major role in slope stability through hydrological and mechanical processes; these effects can be adverse or beneficial to stability (table 1). Hydrological mechanisms that lead to lower pore-water pressures in the soil are beneficial, whereas those that increase pore pressure are adverse. Of the mechanical mechanisms, those that increase shear resistance in the slope are beneficial, whereas those that increase shear stress are adverse (Greenway 1987). Mechanical and hydrological processes have usually been studied separately, but they strongly interact and can even offset one another. Therefore, the two processes need to be considered together when assessing the stabilization of hillslopes by vegetation.

Canopy interception of precipitation—both rain and snow—and subsequent evaporation can prevent considerable amounts of water from reaching the soil surface and may mitigate landslide occurrence. However, surface soil acts as a more significant buffer and tends to dampen pore pressure propagation, and the effect of surface soil probably exceeds most canopy effects during the redistribution of heavy rainfall (Keim and Skaugset 2003). The role of roots in extracting soil water by evapotranspiration is rather limited during cold, wet seasons, when landslides typically occur in temperate regions, whereas in tropical and semitropical climates, such evapotranspiration benefits may be more substantial (Simon and Collison 2002, Sidle et al. 2006).

To reduce subsurface pore-water pressure, efficient subsurface drainage is necessary. Plant roots, alive or dead, can promote slope drainage by functioning as hillslope-scale preferential flow paths that drain subsurface water away from potentially unstable sites. Conversely, when root channels converge or when subsurface flow abruptly terminates in the slope (e.g., dead-end channels), water pressure may concentrate in critical zones in the slope, thus promoting instability. Therefore, flow paths may result in both positive and negative consequences on slope stability. We attempt to answer the question of what the mechanisms and configurations are in which a root or an assemblage of roots plays a positive or a negative role in slope stability. To address this question, precise knowledge of the disposition of roots within the slope is needed, and we suggest that the methodology proposed for root architecture analysis is a suitable approach. First, we describe how roots can actually provide preferential flow paths, then we articulate how root-system architecture can affect preferential flow and improve our insights into slope failure. Finally, implications of vegetation cover for slope stabilization are illustrated by case studies.

How roots create a stable channel when growing in the soil

Water flows through two domains in soil: the soil matrix, consisting of both uniform saturated and unsaturated flow through fine pores, and preferential flow pathways, consisting of single or interconnected macropores (figure 1a).

Table 1. Hydrological and mechanical mechanisms affecting slope stability and their influence on shallow and deep-seated landslides.

Mechanisms affecting slope stability	Type of landslide influenced by the mechanism	
	Shallow, rapid landslides	Deep-seated landslides
Hydrological mechanisms		
Interception of rainfall by canopies	mB	mB
Organic litter increases ground surface roughness and "Biomat flow" (Sidle et al. 2007)	mA (B for runoff)	mA
Roots act as preferential flow paths	B or A	B or A
Roots extract water from the soil	B or mA	B or mA
Mechanical mechanisms		
Roots act as reinforcement fibers in the soil matrix	B	B
Strong roots tie across planes of weakness and anchor soils into the more stable substrate	B	mB
The weight of trees increases the normal and slope-parallel force components	mB or mA	mB or mA
Wind transmits dynamic forces to the soil mantle	mA	mA

A, adverse influence; B, beneficial influence; m, marginal influence.
Source: Adapted from Greenway (1987) and Sidle and Ochiai (2006).

Because of the lack of a consensus definition, we consider *macropores* to be all types of pores with a diameter of more than two millimeters (mm) that support both uniform and nonuniform flow. Various studies have reported macropore diameters ranging from 30 micrometers (Marshall 1959) to many centimeters (Pierson 1983). One way to classify macropores is to distinguish between biotic and nonbiotic macropores. Biotic macropores include root channels and animal burrows, whereas nonbiotic macropores are formed by freeze–thaw and wetting–drying cycles, the dissolution of soil materials, natural soil aggregation, and subsurface erosion (Aubertin 1971). In this article, we deal with biotic macropores and, more specifically, with root channels.

Roots are usually classified as either coarse or fine, with a diameter of 2 mm typically used as the threshold between the two groups (Stokes et al. 2009). In other studies, different criteria have been used to distinguish between coarse and fine roots, such as the ability of fine roots to absorb water and nutrients, whereas coarse roots are more rigid, anchor the plant to the soil, and provide the structural skeleton that supports the fine roots (Zobel 2005).

Initially, fine roots grow in length; this stage is the primary growth stage, which consists of organogenesis and organ length extension. During primary growth, the root apex increases temporarily in diameter (figure 2a) to facilitate penetration through soil material (Bengough et al. 1997). During this process, soil particles may reorganize around the apex surface (figure 2b). Around the thinner portion

of the root that follows the apex, where root hairs appear, soil particles amalgamate with sloughed cells that emanate from the cap of the root apex and with the mucilages emitted by both the root and microorganisms surrounding the root to create the *rhizosheath* (figure 2c). This rhizosheath can be rather large up to 2.5 to 3.0 times the volume of subtending root in monocotyledon root systems—or can even be absent in dicotyledonous root systems (McCully 1995). Because the rhizosheath adheres to the root in dry conditions, this tight mantle around the roots is not the entity that will create a structured channel for preferential flow. The *rhizosphere* is defined as the ensemble composed of the roots, the rhizosheath if there is one, microorganisms, and the soil around the roots (Tarafdar and Jungk 1987), not including bulk soil. The rhizosphere is a focal point in many studies because of its complexity; it exhibits physical, chemical, and biological characteristics that differ from those of rootless soil. The rhizosphere can extend several millimeters beyond the root. For

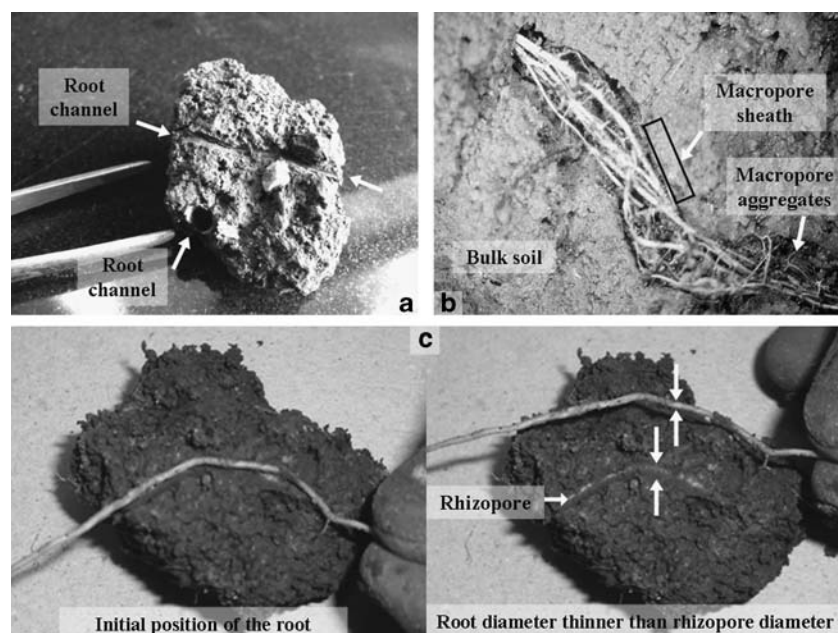


Figure 1. Illustrations of different types of root channels. (a) Two macropores (empty root channels) in silty-textured loam from the Yunnan province, China. (b) Brassica napus roots growing in an existing root channel or biopore. Photograph: Clive E. Pankhurst, Commonwealth Scientific and Industrial Research Organisation Land and Water. (c) The root diameter can be thinner than the rhizopore diameter, leaving a ring-shaped space for potential water flow.

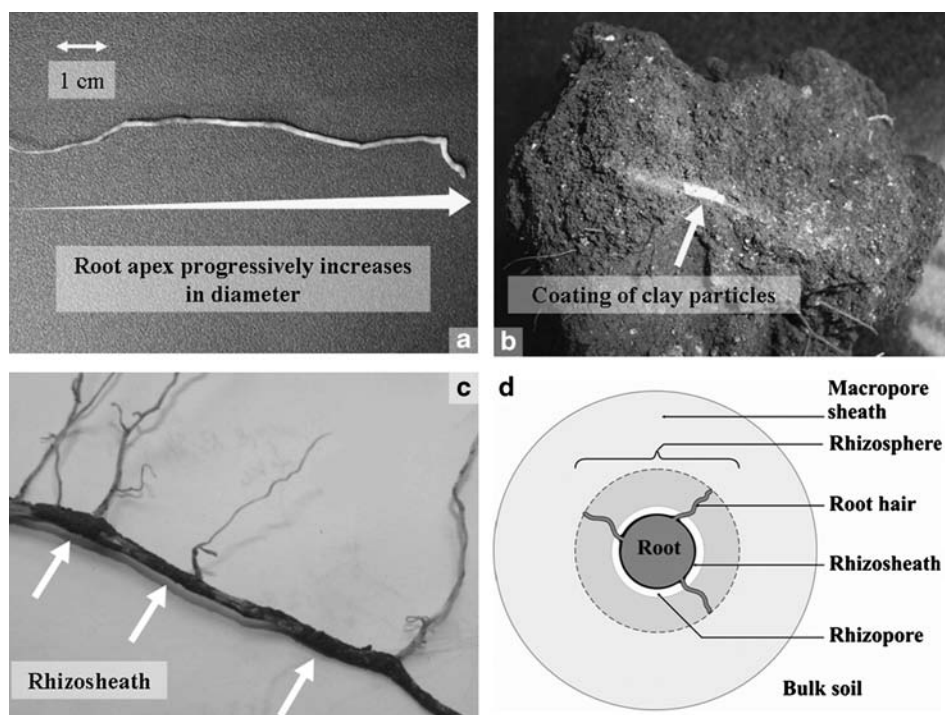


Figure 2. Mechanisms by which fine roots create a stable channel when growing in the soil. (a) As fine roots grow in length, their apex may increase in size to facilitate penetration through soil material. (b) During root penetration, soil particles may reorganize around the root surface, as is shown by this coating of clay particles on the sides of the channel. (c) Behind the root apex, soil particles amalgamate with sloughed cells and mucilages to create the rhizosheath, shown here (white arrows) in an *Agave americana* root. (d) Structure of a macropore, including the rhizosheath, rhizosphere, rhizopore, and pore sheath. Abbreviation: cm, centimeters.

example, soil microorganisms associated with the presence of roots and the enzymes these microorganisms produce can be active within a 4-mm radius around the root (Fageria and Stone 2006). The rhizosphere increases organic matter content around the root, which supports soil aggregation (Fageria and Stone 2006). Fungal mycelia improve the stability of soil aggregates by growing around soil particles and amalgamating them, which increases mechanical stabilization. An increase in aggregate stability will improve channel longevity. Glue-like polysaccharides, which are products of fungal metabolism, also cement soil particles together, which improves chemical stabilization (Tisdall and Oades 1982). Roots themselves release large amounts of organic materials into the soil as the rhizosheath decays and is replaced. Mucilages from roots and microorganisms have the capacity to stabilize soil structure in the rhizosphere by increasing the strength of the bonds among particles.

Mucilages that clog soil aggregates physically decrease the wetting rate, and fungi in the rhizosphere are known to chemically induce water repellency (Hallett et al. 2006). Organic compounds released by roots and microorganisms are hydrophobic below a certain threshold of moisture, but when soils are wet, as is the case when landslides are triggered by rainfall,

these organic compounds are highly hydrophilic (Dekker et al. 1998). Coarse roots do not generally emit exudates, but physical effects may occur that expand the diameter of the root channel through the accumulation of woody tissues, which induces compression forces and rearranges mineral and organic soil particles along the surface of the root (Blevins 1968, Aubertin 1971). The chemical and biological activity in the rhizosphere, together with the physical action of the root, helps establish a relatively stable channel, which could be called a *rhizopore*.

Growing roots may develop inside preexisting pores (figure 1b). For example, in an untilled field of wheat (*Triticum aestivum* L. cv. Janz) that had previously been planted with alfalfa (*Medicago sativa*), 20% (near the surface) to 90% (at depth) of roots grew inside existing pores of clay loam soil (White and Kirkegaard 2010). In some cases, void space may still occupy up to 40% of the macropore vol-

ume, even if the roots tend to stick to the macropore wall along portions of the circumference (Kooistra et al. 1992). In other cases, roots cluster inside a former macropore until up to 10 roots are compressed together (White and Kirkegaard 2010). Even more roots were found in and around macropores than would be expected if their placement were random, which indicates a preferential association between roots and macropores (Stewart et al. 1999). The zone surrounding the rhizosphere, with a higher concentration of roots and a higher microbial biomass density than both the rhizosphere and bulk soil, is called the *macropore sheath* or the *macrosheath* (Pierret et al. 1999; see figure 2d for a summary of terms used in this article). This preferential concentration of roots inside existing macropores and macrosheaths may be due to the easier access to air and water rather than to the low mechanical resistance of these zones. The effects of low mechanical resistance may be beneficial for some species, for which root penetration is retarded in fine-textured soils because of the presence of clay, which acts as a cementing agent. For these species, root growth in or in association with former macropores is more common. Root growth into macropores may also be more common in massive-structured soils (i.e., soils with high bulk density), in which

root penetration is less than that in lower-bulk-density soils. Even if a root is larger in diameter than the channel it encounters, it will be able to enter if the channel walls are weak (Kar and Ghildyal 1975). Roots that grow within the macropore sheath benefit from low soil mechanical resistance and proximity to water, air, and nutrients, but do not experience the drawbacks of growing inside the macropore because the sheath provides greater root–soil contact than the interior of the macropore. Nevertheless, root growth inside the macropore and in the sheath presents other difficulties: (a) Root uptake of low-mobility nutrients (e.g., phosphorus) is restricted because the former root may have previously consumed these nutrients, (b) root clusters can lead to drier soil zones, (c) development of pathogenic fungi is more common, and (d) foraging or grazing soil fauna may be more abundant (Pierret et al. 1999).

In summary, three categories of root channels coexist: channels occupied by the root that made it, empty or nearly empty channels occupied by a decaying root, and channels that were previously empty but that are newly occupied by roots.

How root channels contribute to preferential flow

Although few data exist concerning how root channels contribute to preferential flow, we can induce the likely mechanisms that facilitate such flow paths from root-architecture data and our knowledge of subsurface flow processes. In this article, we use the term *preferential flow pathways* to distinguish those root channels from other pathways that actively transmit water. As was already stated at the beginning of this article, soil wetness depends on infiltration and subsurface flow processes. Water can directly enter macropores that are open to the soil surface. Water moves through soils on the basis of an energy gradient—from high to low energy. Therefore, water does not necessarily flow from higher to lower elevations, although this is generally the case. For water to enter a subsurface macropore and initiate preferential flow, either the entire perimeter or a portion of the macropore should be saturated (or nearly saturated). Therefore, macropores located below a water table or within a limited zone of saturation capture subsurface water and initiate a preferential flow. In contrast, macropores located in unsaturated areas and those surrounded by hydrophobic conditions are bypassed, which causes water to flow around these voids. The network of interconnected macropores is not obvious and continuously evolves—if not physically, at least in its flow activity (Nieber and Sidle 2010).

Numerous investigators have noted water discharging from visible decayed root channels, but water has also been observed to flow from channels occupied by live roots (Aubertin 1971, Noguchi et al. 1997a, Newman et al. 2004). As such, all types of channels associated with roots can act as preferential flow paths and can interact to create networks that potentially include entire hillslopes (Sidle et al. 2001).

The proportion of the total macropore population represented by both live- and dead-root channels is significant,

sometimes up to 70% (Noguchi et al. 1997b) or even up to 100% (Newman et al. 2004) of the macropore population in the upper organic-rich soil layers and up to 35% of the total soil volume (Aubertin 1971). Because the estimated volume of root channels is always greater than the potential volume of water discharged from these channels, it is clear that not all root channels support water drainage (Li and Ghodrati 1994). To our knowledge, no study has segregated live- from dead-root channels with the aim of better understanding preferential flow processes and function.

The root—and its rhizosheath if it has one—evolves within the rhizopore, thereby leaving a ring-shaped or semi-ring-shaped space for potential water flow in response to diameter changes (figure 1c). Fine-root diameter can decrease up to 60% within a diurnal cycle of desiccation at peak afternoon radiation, followed by overnight rehydration (Huck et al. 1970). Roots also retract in volume during dry seasons (Nobel and Cui 1992). Spaces between living roots and their channels can be created near the trunk when windy conditions prevail and the stem bends during wind gusts, which cause the roots to move and displace soil around the channel walls (Hintikka 1972). Such windy conditions principally act on coarse and rigid roots and often precede or accompany rainstorms; therefore, in addition to wind stress (table 1), they can initiate preferential flow along roots starting from the ground surface and may contribute to landslide initiation. Another mechanical effect that can create preferential flow around roots is the tension exerted during minor soil slippage. Such tensile forces may decrease root diameter and might remove part of the bark. This process may cause the root channel to collapse, or it may result in the development of an empty space around the root that allows water to rapidly infiltrate, which causes pore-water pressure to increase in an already mechanically weakened zone. It can be supposed that a series of such microfailures can in turn trigger a larger landslide (Chigira 2001).

The space between the root and the soil can be detrimental to plant fitness, because the root's uptake of water and nutrients may be limited, or it can be beneficial, as in very dry conditions in which the air occupying the space acts as a buffer and prevents the soil from extracting water from the root. Van Noordwijk and colleagues (1992) showed that root hairs are more numerous when spaces exist between the root and the soil. Root hairs may not be tightly bound to the soil because they do not penetrate the bulk soil (Kooistra et al. 1992) and because they do not increase root anchorage (Bailey et al. 2002).

Research on channels occupied by decaying roots relies on a better understanding of root necrosis and root longevity. Studies in which minirhizotrons (microvideo cameras) were used to monitor growth and in which isotope-based estimates of carbon residence times in fine roots were used showed that the life spans of various roots can vary from several days to five or six years (Strand et al. 2008). Assessing root decay and regrowth is difficult because turnover depends on many parameters, such as the plant species, root

thickness, age, and environment. The decay and regrowth rates have often been modeled as time-dependant functions (Burroughs and Thomas 1977) correlated with plant species, root densities in the soil, and, at the root scale, with the functional category of the root (Atger and Edelin 1994). Root resistance to decay was also found to be higher in resinous roots than in nonresinous roots (Ziemer and Swanston 1977). Therefore, preferential flow networks will develop more slowly after the death of members of a species that have a high resin content in their roots (e.g., conifers). At the onset of senescence, the root channel is filled with organic material derived from the root itself. Depending on the species, the type of root, and the climate, the bark can disappear first or can remain and contribute to the integrity of the channel. If the rhizopore remains empty, clay particles may first deposit as skins on the sides of the channel (figure 1b; Aubertin 1971). With time, other particles crumble into the channel if they are eroded from the interior through sapping caused by erosion and abrasion during subsurface sediment transport. Thus, seepage forces enlarge the channel but may eventually obstruct it. Roots that occupy an old rhizopore can be initiated by the mother root system or by neighboring root systems. These dead-root channels thus provide interconnections among root systems.

Therefore, to understand how preferential flow develops and evolves with time, it is important to evaluate root emission (i.e., the birth of new roots) and root decay, not only in a quantitative way—by specifying decay and emission rates—but also in terms of location, the angle of insertion, and the rate of growth within a single root system.

Root architecture as an approach to understanding preferential flow

As far back as the eighteenth century, when scientists and naturalists began looking at the morphological characteristics of plants, they noticed that some of these features were inherent to a given species and could therefore be useful in the process of classifying species. It was not until the 1970s that the study of plant architecture was formalized, with rules to describe the equilibrium between endogenous growth processes and the exogenous constraints exerted by the environment on plant development at any given time (Hallé and Oldeman 1970). The aim of architectural analysis is to identify and understand these endogenous processes and to separate them from the plasticity of their expression that results from external influences. Architectural analysis is thus a tool to describe the progressive development, or *ontogeny*, of a given species. This type of analysis takes into account that a plant is a modular organism composed of several units. Recently, more emphasis has been placed on the implications of plant architectural analysis for agriculture and crop yield improvement, as well as on commercial forestry and the urban planning associated with tree growth and health (Barthélémy and Caraglio 2007).

Roots provide a weak visual component of plant architecture because of the difficulties in observing them and

their indistinct morphological characteristics. On the basis of crude observations of the overall shape of root systems or of simple morphological classifications such as “lateral” or “vertical” roots, the conceptualization of architectural models for different root systems has been difficult. In more recent studies, the ontogeny of root systems has been investigated. For example, Atger and Edelin (1994) described successive architectural patterns at different physiological stages of root-system formation. Khuder and colleagues (2007) demonstrated the reiteration of architectural patterns within root systems. Functional properties have been used to define different categories of roots. This approach first used the term *root functional architecture* (Zobel 2005). For example, the ability to uptake water and nutrients varies from one root to another and even along individual roots. A root with a high specific root length to root mass density (SRL:RMD) ratio would be better equipped to scavenge nutrients in the soil, whereas a low SRL:RMD ratio would characterize a root in which the plant invests more resources for mechanical support (Roumet et al. 2006). To better describe root functional architecture, topological analysis (Fitter 1985) can be used to quantify branching patterns (i.e., how roots are geometrically linked). The potential effects of different branching patterns on slope stability were summarized by Stokes and colleagues (2009). The architectural analytical methods to be used must be chosen carefully, in accordance with the traits to be quantified (Danjon and Reubens 2007). Excavation of the entire root system, root counts on profile walls, measurements of windthrown trees, and soil coring can provide data to support empirical and numerical modeling. Predictive models are also useful for estimating root densities and predicting root growth (Pierret et al. 2007) without time-consuming excavation.

How different root architectures can influence preferential flow paths

The architectural analysis of plant root systems growing on hillslopes can help us understand the role that roots play in preferential flow. Root traits (e.g., diameter, length, sinuosity, decay rate, orientation, topology) partially determine root-channel development and thus have an impact on preferential flow. In general, longer and wider root channels more efficiently route water away from unstable zones; however, some root orientations may be more efficient than others. Roots oriented downslope convey water more efficiently (figure 3a). Depending on their orientation, large root extremities can represent dead-end paths for water flow, thus leading to a local increase in water pressure (figure 3b). The collapse of root channels acts in a similar manner.

Depending on their orientation and position with regard to the stem, forks may divide or concentrate flow. Zones of water concentration where root forks converge downslope may increase water pressure (figure 3c). Forks can also occur as the result of interconnections among roots from different plants.

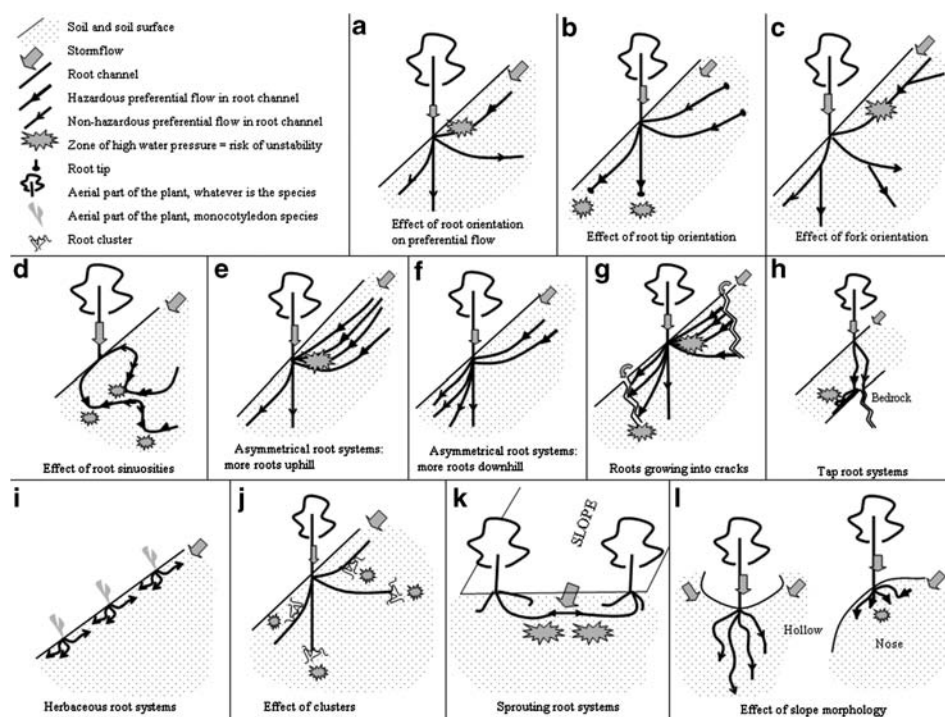


Figure 3. Illustrations of different scenarios of the effects of root architecture on preferential flow. (a) Downslope root orientation is more efficient for transporting excess water. (b) Root extremities can represent dead-end paths for water flow. (c) Root branching may divide or concentrate flow. (d) Sinuous roots divert or concentrate water fluxes, depending on their orientation. (e) When the majority of roots grow upslope, there is a potential for pore-water accretion at shallow depths to occur near the plant stem. (f) When more roots grow downslope, drainage of water is facilitated. (g) Roots growing toward cracks enhance the risk of water accretion. (h) Taproot systems convey water to deeper soil, where it can drain into cracks in the bedrock if cracks are present. However, if the soil–bedrock limit is impermeable, zones of high water pressures can be created. (i) Tuft-root systems allow water to infiltrate into upper soil layers. (j) Clusters of roots act as sponge-like structures and concentrate high water pressures. (k) Roots oriented perpendicular to the slope gradient capture downward water flow. (l) Topographic situations can combine various water-flow and root-distribution patterns.

Sinuous roots are variants of forks: Depending on their orientation and location, they may either divert or concentrate water fluxes. In situations in which root curvature concentrates fluxes, high pore-water pressure can induce unstable zones in the soil (figure 3d).

Root systems that exhibit preferential growth upslope or downslope need special consideration. From a strictly mechanical point of view, the orientation in which roots are aligned with regard to slope direction can affect soil reinforcement, depending on whether the plant is located at the top or at the bottom of a slope (Genet et al. 2010, Thomas and Pollen-Bankhead 2010). From a hydrological perspective, root orientation is also important. In some systems, more roots are oriented downslope than upslope (Stokes et al. 2009). This preferential gravitropism depends on the species, on nutrients, and on the soil's physical properties. If there are

more roots growing downslope, drainage of water is facilitated in this direction (figure 3f). That some systems have more roots growing upslope from the stem suggests that some roots may undergo hydrotropism rather than gravitropism if more water is located upslope. Roots do not search for water actively but are able to reorient or branch when they reach a moist patch. Systems with more roots growing upslope will drain water less efficiently and will promote pore-water accretion at shallow depths near the main stem (figure 3e).

Depending on the climate, dry seasons can correspond to hot or cold temperatures. In the former case (i.e., dry, hot conditions), cracks in the soil can appear, caused by the swelling and shrinkage of clay minerals. In colder conditions, cracks occur because of the freeze–thaw process. Roots can proliferate in cracks that regularly accumulate water during even short rainfall. During heavy rain, water flow in cracks increases, and roots may even become asphyxiated and die. The growth characteristics of roots in soil cracks can therefore have consequences for the stability of slopes (figure 3g).

Structural cracks also exist in bedrock. If roots grow through the soil and are able to penetrate the bedrock and fissures in the bedrock, the water they convey may dissipate inside the bedrock. Such a pathway may reduce pore-water pressure accretion in the soil but could transfer such pore pressures into the bedrock and contribute to rockfall. In contrast, if roots do not grow into the bedrock, a layer of high root density may develop above the bedrock, which may induce locally high pore pressures and which could trigger landslides along this potential slip surface (figure 3h). Many inherently deep-rooting species can be found in thin soils over bedrock (Stokes et al. 2009). Deep-rooting species can benefit from preferential flow, because flow may short-circuit the upper soil layers. Therefore, deep-rooted coppiced plants, with their root systems already in place, can grow more rapidly and more vigorously than newly planted individuals, even when the latter are fertilized.

The effects of tuft root systems, which are typical of herb species, on water pathways are unclear. Inhibiting or diffusing water flow at depth near potential slip surfaces will help promote the stability of slopes. However, in reality, the soil surface is not homogeneous, and a single crack can concentrate subsurface flow. Therefore, fibrous (figure 4a) and shallow root systems dispersed along the slope may act as preferential flow paths that dissipate pore-water pressure at a given point (see the upslope portion of figure 3i).

Clusters of fine roots are sometimes observed along or at the end of coarse roots and correspond to zones of major nutrient and water uptake. Fine roots have high decay and emission rates, and clusters may therefore manifest as sponge-like structures. These structures may soak up water during heavy rains and may contribute to high-water-pressure nodes, especially at dead-end flow paths (figure 3j).

In certain species, superficial underground creeping stems (rhizomes) or roots ensure asexual reproduction. These organs link two or more aboveground vegetative structures together. If they are oriented perpendicular to the slope, such structures may intercept downslope water flow, which could lead to a local increase in water pressure at the zone of interception, which, in turn, could result in a locally fragile zone on the slope (figure 3k). As the plant community matures, these structures may well develop into a dense root mat.

Topography is also important with regard to subsurface water movement and slope stability. Concave slope segments (hollows) promote the convergence of subsurface flow (Tsukamoto 1987). However, some studies indicate that roots in hollows are more evenly distributed in the soil

column than those on convex slopes where roots concentrate on the upper horizons of soil (Hales et al. 2009). Therefore, in hollows, water flows may be diverted more efficiently toward deeper soil (figure 3l). As such, root architecture, along with its interaction with water, nutrients, and topography, can affect slope stability.

Case studies of how root-system architecture influences subsurface flow

Examples in which the interactions among root architecture, preferential flow, and landslides have been examined are few. One of the earliest studies was by Gaiser (1952), who hypothesized that root channels must serve as large openings for rapid water flow. He therefore mapped the presence of cone-shaped channels formed from decayed roots through the A, B, and C horizons of a temperate forest soil. The forest was dominated by white oak (*Quercus alba*), with black walnut (*Juglans nigra*), hickory (*Carya* spp.), and dogwood (*Cornus* sp.) present in smaller quantities. The shrubby dogwood possesses a highly branched superficial root system, whereas the three tree species generally have much deeper root systems (Kutschera and Lichtenegger 2002). In particular, white oaks usually possess a taproot system when they are young, with a single large, vertical central root and thinner, less numerous lateral and sinker roots (figure 4b). But when white oaks mature, their taproot growth decreases, and the proportion of lateral, oblique, and sinker roots increase, forming a heart-shaped root architecture (figure 4c; Köstler et al. 1968). Except for one 90-year-old tree, the white oaks examined by Gaiser (1952) were less than 20 years old. Therefore, the cone-shaped vertical channels, which decreased from 36 to 5 centimeters (cm) in diameter at the soil surface to 3 to 0 cm in diameter at the furthest extremity, were likely formed by the taproots of juvenile white oaks and allowed for an increased infiltration of water to deeper soil (figure 3h). Gaiser (1952) concluded that further research on subsurface flow in forest soils should not neglect the effect of root-induced macroporosity in the soil profile. In a similar type of forest, Whipkey (1969) measured subsurface flow after rainfall simulations. Through observations in trenches, Whipkey (1969) observed water leaking from root channels, with the greatest flow from channels just beneath or at an angle to the area where the simulations were carried out. The vegetation was composed principally of species with heart-shaped root systems (figure 4c), such as beech (*Fagus* sp.) and mature oaks (*Quercus* sp.), as well as species with more superficial plate-shaped root systems (figure 4e; Köstler et al. 1968) consisting of large or numerous lateral roots such as ash (*Fraxinus* sp.) and cherry (*Prunus* sp.). This mixture of species with different rooting patterns, with thick lateral roots in the proximity of vertical and oblique roots, can explain the high interconnection of water flow observed by Whipkey (1969). Similarly, Newman and colleagues (2004) observed lateral subsurface flow resulting from a melting snowdrift in a semiarid ponderosa pine (*Pinus ponderosa*) forest. All lateral subsurface flow occurred in the B horizon

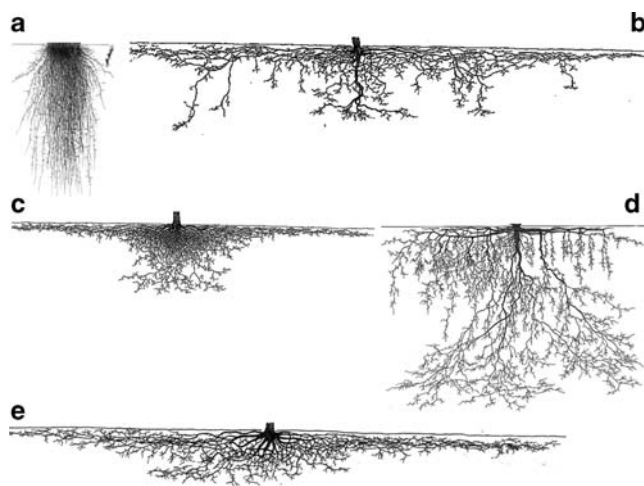


Figure 4. Five basic types of root systems. (a) Tuft root systems typical of *Poaceae*. (b) Taproot system. (c) Heart-shaped root system. (d) Root system with a large taproot and large horizontal lateral roots from which emerge vertical sinkers. (e) Plate-shaped root system. Sources: Panels (a) and (d) are reprinted from Kutschera and Sobotik (1997) with permission from Landesmuseum. Panels (b), (c), and (e) are reprinted from Kutschera and Lichtenegger (2002) with permission from Stocker.

and originated from macropores occupied by decaying or live roots. Only a few lateral roots were found in the C horizon. When soil conditions are not limiting, the ponderosa pine possesses a single long taproot with superficial wide-spreading lateral roots (figure 4d; Stoecklein 2001); therefore, subsurface flow along roots and channels occur mainly in the shallower soil layers.

Lateral subsurface flow was also observed by Noguchi and colleagues (1997a) in a Malaysian tropical rainforest dominated by *Koompassia malaccensis*, *Eugenia* spp., and *Canarium* spp. Noguchi and colleagues (1997a) found relatively high hydraulic conductivities in the upper layers of soil. Dyed water dispersed in the upper horizon and then percolated through vertical and downslope-oriented root channels. The root-system architectural type of *K. malaccensis* is not known, but *Eugenia* spp. and *Canarium* spp. generally possess roots mostly in the upper organic-rich layer of soil (Nishimura and Suzuki 2001, Ghani et al. 2009). This layer creates a porous zone in which high water pressures can develop from both saturated and nonsaturated flows. Noguchi and colleagues (1997b) also highlighted the similarity between the direction of subsurface flow and the direction of root growth in the upper layers of a semitropical forest dominated by *Cryptomeria japonica* and *Chamaecyparis obtusa*, two species with shallower, more plate-like root systems (figure 4e) that expand more on the uphill side of the stem (figure 3e). In deeper soil horizons, the direction of flow did not originate from root channels, because these were not numerous at that depth, but from soil cracks and weathered bedrock. Also in Malaysia, Collison and Anderson (1996) used a modeling approach applied to vegetation and soil data originating from long slopes. It was found that the roots of the tree species present did not usually cross the potential shear surface when they were growing in the middle of slopes with deep soils. Higher water pressures also resulted at the base of the rooting zone or even below this zone. Collison and Anderson's (1996) advice was that on such long slopes with deep soils, greater mechanical integrity will occur when trees are planted at the toe of the slope, where the potential shear surface is more likely to cross the rooting zone. Farther up the slope, grasses should be planted to reduce rainwater infiltration (figures 3i, 4a).

Although most case studies have been performed in forests, Cammeraat and colleagues (2005) studied hydrological processes on agricultural terraces on which rain-fed orchards—mainly cherries (*Prunus avium*), olives (*Olea europaea*), and almonds (*Prunus dulcis*)—were cultivated. These terraces in southern Spain were abandoned over 50 years ago and have since been colonized by shrubs such as gorse (*Ulex parviflorus*) and hawthorn (*Crataegus monogyna* Jacq.) and by Aleppo pine (*Pinus halepensis*), which has a taproot with long laterals and sinkers (figure 4d). Through a series of rainfall simulations, Cammeraat and colleagues (2005) observed that wetting fronts were more irregular and that percolation was deeper in the terraces that had been abandoned the longest. The increase in landslide

occurrences was attributed to the faster transfer of rainfall to the potential slip plane by root channels. Although hawthorn has a shallow and wide-spreading root structure (figure 4e), which is probably not responsible for major preferential flow, stems can sprout directly from the lateral roots (figure 3k; Kutschera and Lichtenegger 2002). Therefore, juvenile plants can have relatively large and shallow roots that run continuously among stems underground. This type of root system can enhance lateral subsurface flow. Aleppo pine possesses the large vertical taproot and thinner lateral roots typical of *Pinus* spp. (figure 4d); therefore, flow is likely to be greater along the thick taproot. This preferential flow along the thick taproot can create zones of high water pressure and, therefore, fragility at depth.

Species possessing superficial underground running stems and rhizomes (figure 3k) can also increase the likelihood of high pore pressures occurring just beneath the soil surface after heavy precipitation. Although Stokes and colleagues (2007) attributed the greater number of shallow landslides that occur in big-node bamboo (*Phyllostachys nidularia*) forests to poor root-system anchorage and superficial rhizomes that rarely traversed the potential shear surface, it is probable that dense root mats of rhizomatous bamboo species also have consequences for hydrological processes. Ide and colleagues (2010) found that during rainfall simulations in moso bamboo (*Phyllostachys pubescens*) forests, a high density of roots and rhizomes in the surface soil permitted a greater portion of rainfall to infiltrate into the soil. During heavy rain events, the dense root mats and thick rhizomes could even change the direction of rain flow, which resulted in lateral preferential flow. To a certain degree, this lateral flow ran out of the partially exposed roots and contributed to moderate amounts of surface runoff.

Even in peatland ecosystems, root channels can create discontinuities in the soil profile. Holden (2005) found that root channels of heather (*Calluna* spp.) increased the presence of macropores and the occurrence of bypass flow compared with cotton grass (*Eriophorum* spp.) and peat mosses (*Sphagnum* spp.) when they were growing on blanket peats. This increased flow was due to differences in the root architectures of these species: Heather possesses a woody and highly branched root system, whereas cotton grass has a prominent bulbous tussock root system composed of fine roots (figures 3i, 4a). Sphagnum moss cushions have no roots; only decaying litter material concentrates underneath these mosses.

One further highly complex environmental setting in which root architecture can interact directly and instantaneously with several different hydrological processes is a riverbank. To ensure the mechanical integrity of a riverbank, vegetation strips are widely planted by river managers, and careful consideration of what species to plant and how they should be planted is required. Simon and Collison (2002) gathered geotechnical, matrix suction, and pore-water-pressure data from streambank plots under three riparian covers: bare soil, eastern gamma grass (*Tripsacum dactyloides*) cover, and a mixture of mature trees composed

of *Platanus occidentalis*, *Liquidambar styroflora*, and *Betula nigra*. These authors demonstrated that after heavy rains, soil in plots with tree cover had positive water pressures, which counterbalanced the mechanical reinforcement of roots and could be detrimental to riverbank stability. The perennial eastern gamma grass has a superficial root system composed of numerous well-developed rhizomes and roots growing to a depth of 20 cm. Even if the concentration of root biomass was higher for eastern gamma grass than for the three tree species present on the plots, the roots of the tree species grew to a depth of over 80 cm. Preferential flow was thus initiated along these deep roots, and positive water pressures were created. Simon and Collison (2002) therefore demonstrated the potential for grasses to be used as effectively as trees to increase stream bank stability.

Conclusions

Because they have specific properties, organization, kinetics, and control factors, root channels represent a specific type of macropore. These characteristics of root channels need to be considered with reference to their influence on preferential flow. Root architecture analysis facilitates the upscaling of processes that occur at the individual plant root scale to processes that occur at the hillslope scale. Knowledge of root architecture at the hillslope scale is an important factor in addressing the complex nature of preferential flow, which has been demonstrated to influence landslide initiation (Sidle et al. 2001). Live and dead components of root systems can either promote or dissipate soil water pressure, depending on configuration, orientation, and interconnections; therefore they may either enhance or decrease the

potential for shallow landslides during storms. Although ecological engineers are beginning to understand the benefits of the mechanical reinforcement of rooting systems, the potential effects of roots on subsurface flow should be given more consideration. As has been discussed in this article, several studies have made noticeable advances in the understanding of the effects of root channels on subsurface flow, but further research is still needed in order to better characterize the interactions among vegetation cover, root architecture, and the initiation of bypass flow. Some ideas for further research are developed in box 1. Such knowledge could significantly advance our insights into why certain hillslope segments fail during storms, whereas other segments remain stable.

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Box 1. Further research to better understand the effects of root channels on slope stability.

Studies to better understand the influence of root channels on slope stability are few, and slope failures often remain poorly understood. More focused research and suitable techniques are needed, especially concerning the geometry of the hydraulic passages provided by interconnected root channels. We need to consider the specific distribution of preferential flow paths, rather than the general influence of root systems and root channels on hydrological processes. Using tensiometers, Whipkey (1969) showed no appreciable change in hydraulic head at any time within the soil matrix. This result supports the observation that lateral flow was not moving as interflow through the general soil matrix but was moving, rather, through bypass channels. It is therefore unlikely that measurements of soil water pressures by tensiometers can demonstrate the existence of preferential flow paths. Dye tracers are probably one of the best ways to follow water movements in soils (see Flury and Wai 2003 for a description of useful dye tracers). Tracer-highlighted channels (particularly root channels) can then be observed *in situ* on trench walls and other excavations. X-ray-transmission-computed tomography, as well as resin and plaster impregnation, can also be used to analyze channel architecture in cores or blocks of undisturbed soil (Noguchi et al. 1997b).

In a field situation, parameters such as the presence of soil structural macropores, the homogeneity of the initial water content of the soil or the homogeneity of the bulk density of the soil are not easily controllable and can hamper the study of preferential flow through root channels. Therefore, one possibility would be to combine the use of tracers with laboratory measurements of soil shear in a large-scale Casagrande box. Real plants could be grown directly in shear boxes with controlled conditions of soil structure, soil texture, water content, and so on. Rainfall simulations in which stained water is used would allow for active channels to be identified on nonsheared control boxes and compared with active channels in sheared boxes.

Any step forward in our knowledge of root channels, such as knowledge of root-channel density, interconnectivity, and permeability, can bring useful insights for preferential flow modeling. A usual way to model preferential flow in soils is to consider a two-phase milieu: soil matrix and macropores. Although matrix flow is well known, macropore flow and, more particularly, root-channel flow still needs to be better characterized. The evolution of root-channel flow characteristics in response to temperature or precipitation changes could also be useful for improving the modeling of climate change impacts (Collison et al. 2000).

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Murielle Ghestem (murielle.ghestim@cirad.fr) is a PhD student at Agro-ParisTech College, France. Alexia Stokes is a scientist with the French National Institute for Agricultural Research. They both work at the joint research unit Botany and Computational Plant Architecture (AMAP), Montpellier, France. Roy C. Sidle is the director of the Ecosystems Research Division of the US Environmental Protection Agency in Athens, Georgia.



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